

Solutions to exercises in Lecture 10

Q1) substituting the equations, we get

$$u_i^{n+1} = u_i^n + \frac{\Delta t}{\Delta x} \left(c u_{i-1}^n + \frac{1}{2} c \left(1 - \frac{\Delta t}{\Delta x} c \right) [u_i^n - u_{i-1}^n] - c u_i^n - \frac{1}{2} c \left(1 - \frac{\Delta t}{\Delta x} c \right) [u_{i+1}^n - u_i^n] \right)$$

$$u_i^{n+1} = u_i^n + c \frac{\Delta t}{\Delta x} \left(u_{i-1}^n + \frac{1}{2} (u_i^n - u_{i-1}^n) - u_i^n - \frac{1}{2} (u_{i+1}^n - u_i^n) \right) + c^2 \frac{\Delta t^2}{\Delta x^2} \left(-\frac{1}{2} (u_i^n - u_{i-1}^n) + \frac{1}{2} (u_{i+1}^n - u_i^n) \right)$$

$$u_i^{n+1} = u_i^n + c \frac{\Delta t}{\Delta x} \left(\frac{1}{2} u_{i-1}^n - \frac{1}{2} u_{i+1}^n \right) + c^2 \frac{\Delta t^2}{\Delta x^2} \left(\frac{1}{2} u_{i-1}^n - u_i^n + \frac{1}{2} u_{i+1}^n \right)$$

$$u_i^{n+1} = u_i^n - c \Delta t \left(\frac{u_{i+1}^n - u_{i-1}^n}{2 \Delta x} \right) + \frac{c^2 \Delta t^2}{2} \left(\frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2} \right)$$

as expected for Lax-Wendroff scheme

Q2) Multiplying $\hat{L}_x(\Delta t) \hat{L}_y(\Delta t)$ we get

$$\left(1 - c_x \Delta t \partial_x + \frac{c_x^2 \Delta t^2}{2} \partial_{xx} \right) \left(1 - c_y \Delta t \partial_y + \frac{c_y^2 \Delta t^2}{2} \partial_{yy} \right)$$

$$= \left(1 - \Delta t (c_x \partial_x + c_y \partial_y) + \frac{\Delta t^2}{2} (c_x^2 \partial_{xx} + 2 c_x c_y \partial_x \partial_y + c_y^2 \partial_{yy}) \right) + \text{higher order terms in } \Delta t, \text{ such as } \Delta t^3 \text{ \& } \Delta t^4$$

Comparing the above to Eq. (13) in the lecture note, the key difference is:

$$c_x c_y \partial_x \partial_y + c_y c_x \partial_y \partial_x \quad \text{vs} \quad 2 c_x c_y \partial_x \partial_y$$

If $\partial_x \partial_y$ commute with $\partial_y \partial_x$ then the two expressions are the same.